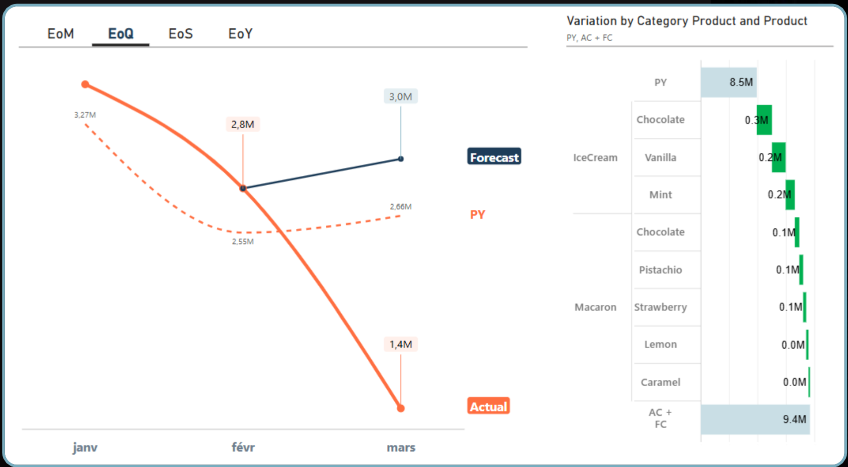
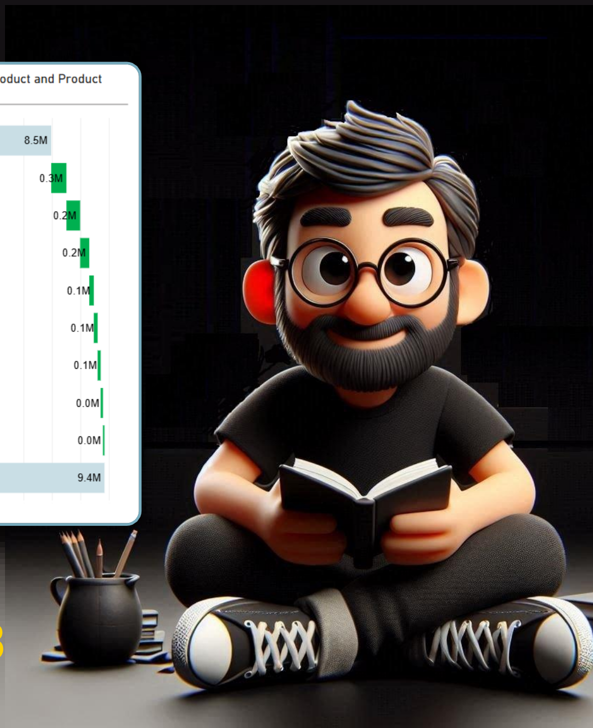
Use case with PowerBI

"Finance Toolbox"

FREE
PowerBI pbix
to practice

End of Period forecast
(by Naïves Approach)





Patou Tips #31 ▶ Toolbox #3

About this Toolbox and datas…

1. **Try the version online**
   Toolbox #2 PowerBI online

2. **Practice**
   Find all the resources on the Patou Tips GitHub
   Link GitHub to resources Toolbox #2

3. Or directly on the project from the "**GitHub button**" at project top left.

About this Toolbox and datas…

The dataset comes from the fictitious Parisian company "IceCream' Macaron," taken from my book "Financial Forecast with PowerBI: Story of a Point."

In this Toolbox, we want to project sales in the future by generating forecast with the CAGR method

Learn more with my book "Story of a Point" Financial forecast with PowerBi (available in dec 2025).

Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

Use case with PowerBI

"Finance Toolbox"

End of Period forecast (by Naïves Approach)

1 When and how to use it?

2 How it works?

3 About modeling...

4 About DAX measures...

5 About visualizations...

1

When and how to use CAGR method?
Features of this forecast (1/2)

The CAGR (Compound Annual Growth Rate) is a financial concepts used to estimate the performance of an investment or finally like in the toolbox, the performance of sales.

Typical applications	In the financial field, it is usual to make an estimate after the 15th of each month in order to provide management with probable economic information at the end of the month or period. this is called Cut-off and perhaps on financial data or activity volume data. These methods can also be useful for end-of-year estimates such as Q3F or reforecasting.
General type	Time Series Analysis
Drivers used (1)	Trend
Data required (Reference period)	A minimum of 3 values of sales history for a period. A period could be years, months or days…
Projection	3-period forecast. Short terms
Identification of the turning point	Poor
Pros	Simple statistical models

Use case with PowerBI

"Finance Toolbox"

(1) More informatiion with my book about forecast drivers.

Performance of an investment (with CAGR)

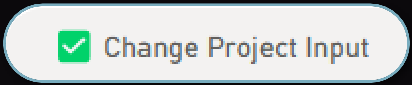
1

When and how to use CAGR method?
Features of this forecast (2/2)

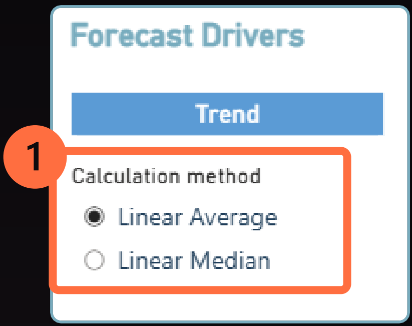
Cons	Does not take into account the seasonal effect. Use only for product on maturity step of the “Product life cycle”.
Note	Please note, for the median calculation, the values must be sorted in ascending order.
Forecast calculations	<u>Linear Average</u> = AVERAGE (Value with Sales from a selected range of date) <u>Linear Median</u> = MEDIAN (Value with Sales from a selected range of date)

2

How it works?



The button "Change Project Input" allows you to configure your simulation



1 Calculation method.

2 Forecast extension period.

Use case with PowerBI

"Finance Toolbox"

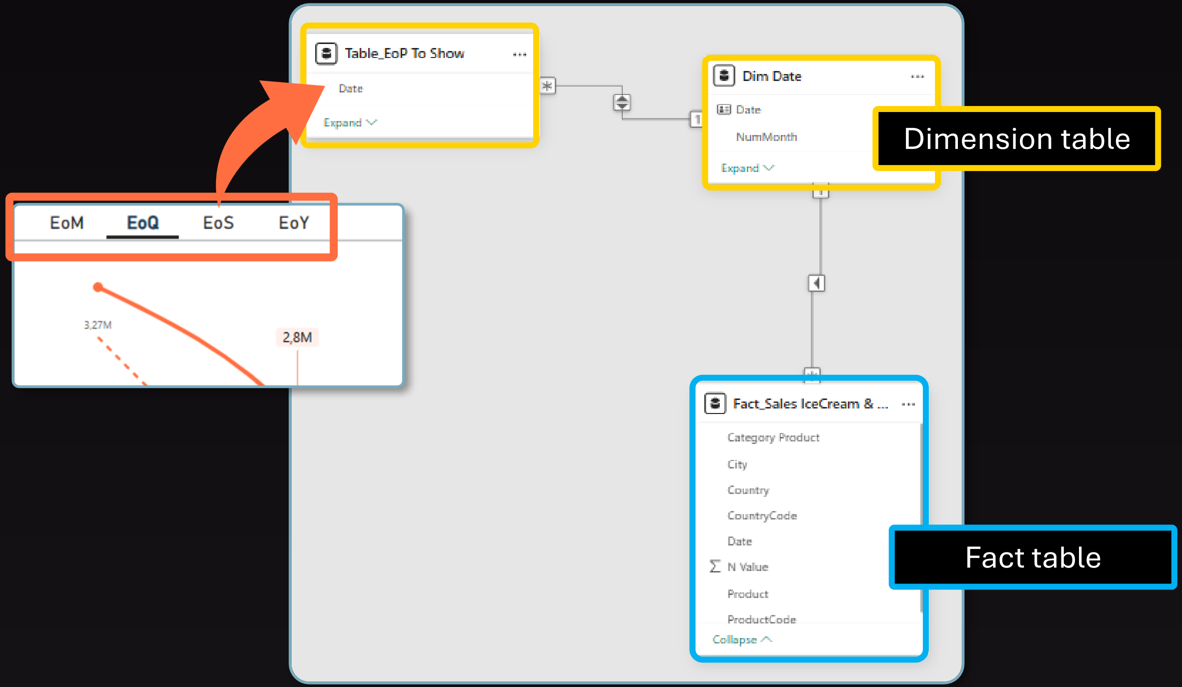
Performance of an investment (with CAGR)

3

About modeling
Main architecture...

Star schema

The “N value”(1) is in the heart of the project!



(1) See Patou Tips # 25 “The complete guide to start with PowerBI” for more information.

4

About DAX measures…
Measures organization

- @Sales IceCream & Macaron
  - Actual

```
1 Actual =
2 SUM('Fact_Sales Data'[N Value])
```

- TB_003 EoP Forecast
  - Calculation
    - AC (for graph) (TB_003)
    - AC + FC (TB_003)
    - FC (for graph) (TB_003)
    - Δ PY (%) (TB_003)
    - Δ PY (TB_003)
  - Visualization
    - Base line for graph (TB_003)
    - Calculation Method Selected (TB_003)
    - Days_Rate (TB_003)
    - EoP Selected (TB_003)
    - Label Forecast Calculation Period (TB_003)
    - Label Forecast Period To Extand (TB_003)
    - Method usage message (TB_003)

The sales measure contains the "N Value", this is the pivot measure.

All measurements are organized into two main folders:

- Calculation: Measurements required for end-of-production forecast calculations.
- Visualization: Measurements used to build visualizations.

Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

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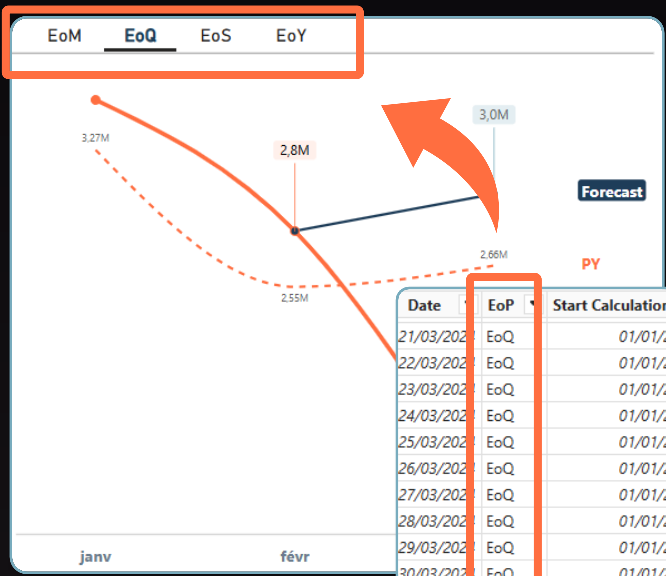
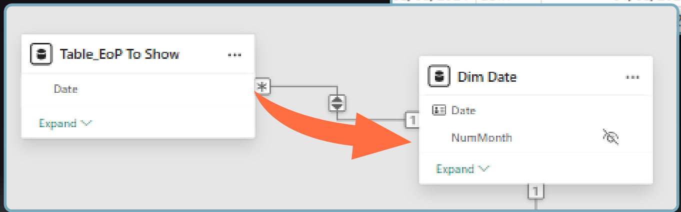
About DAX measures...
Measure details

« Table_EoP To Show » measure



Date	EoP	Start Calculation	End Calculation/Start Forecast	End Forecast	Id_EoP	EoP Description
21/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
22/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
23/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
24/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
25/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
26/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
27/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
28/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
29/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
30/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
31/03/202	EoQ	01/01/2024	15/03/2024	31/03/2024	2	EoQ (End of Quarter)
01/03/202	EoM	01/03/2024	15/03/2024	31/03/2024	1	EoM (End of Month)
02/03/2024	EoM	01/03/2024	15/03/2024	31/03/2024	1	EoM (End of Month)
		[illegible]	15/03/2024	31/03/2024	1	EoM (End of Month)

The first step is to generate a date table that will be used by a slicer. With a single click, the dates in the "Dim Date" will be activated to not only display the desired date ranges but also to indicate which date range to consider when calculating the forecast.



This table is created with DAX code, then linked in modeling with the "Dim Date".

Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

4

About DAX measures…
Measure details

« Table_EoP To Show » measure (1/2)

```
Table_EoP To Show =
VAR Last_Sales_Day = CALCULATE(MAX('Fact_Sales IceCream & Macaron'[Date]),REMOVEFILTERS())
VAR Selected_Semester = CALCULATE(SELECTEDVALUE('Dim Date'[Semester]),'Dim Date'[Date]= Last_Sales_Day)
VAR Selected_Year = CALCULATE(SELECTEDVALUE('Dim Date'[Year]),'Dim Date'[Date]= Last_Sales_Day)

VAR Start_Of_Month =
    CALCULATE(
        STARTOFMONTH('Dim Date'[Date]),
        'Dim Date'[Date]= Last_Sales_Day)

VAR End_Of_Month =
    CALCULATE(
        ENDOFMONTH('Dim Date'[Date]),
        'Dim Date'[Date]= Last_Sales_Day)

VAR Start_Of_Quarter =
    CALCULATE(
        STARTOFQUARTER('Dim Date'[Date]),
        'Dim Date'[Date]= Last_Sales_Day)

VAR End_Of_Quarter =
    CALCULATE(
        ENDOFQUARTER('Dim Date'[Date]),
        'Dim Date'[Date]= Last_Sales_Day)

VAR Start_Of_Semester =
    CALCULATE(
        FIRSTDATE('Dim Date'[Date]),
        FILTER('Dim Date',
        'Dim Date'[Semester]= Selected_Semester && 'Dim Date'[Year]=Selected_Year))

VAR End_Of_Semester =
    CALCULATE(
        LASTDATE('Dim Date'[Date]),
        FILTER('Dim Date',
        'Dim Date'[Semester]= Selected_Semester && 'Dim Date'[Year]=Selected_Year))
```

For the **first part of the formula** we create the variables which are the starting and ending points of the dates to be shown and the dates to be considered for the calculation of the forecast

4

About DAX measures...
Measure details

« Table_EoP To Show » measure (2/2)

```
VAR Date_To_Show =
UNION(
    ADDCOLUMNS(
        DATESBETWEEN('Dim Date'[Date],Start_Of_Month,End_Of_Month),
        "EoP","EoM",
        "EoP Description","EoM (End of Month)",
        "Id_EoP",1,
        "Start Calculation",Start_Of_Month,
        "End Calculation/Start Forecast", Last_Sales_Day,
        "End Forecast", End_Of_Month),

    ADDCOLUMNS(
        DATESBETWEEN('Dim Date'[Date],Start_Of_Quarter,End_Of_Quarter),
        "EoP","EoQ",
        "EoP Description","EoQ (End of Quarter)",
        "Id_EoP",2,
        "Start Calculation",Start_Of_Quarter,
        "End Calculation/Start Forecast", Last_Sales_Day,
        "End Forecast", End_Of_Quarter),

    ADDCOLUMNS(
        DATESBETWEEN('Dim Date'[Date],Start_Of_Semester,End_Of_Semester),
        "EoP","EoS",
        "EoP Description","EoS (End of Semester)",
        "Id_EoP",3,
        "Start Calculation",Start_Of_Semester,
```

For **the second part of the formula**, the table data is created for each "EoP". This data will be useful for graphs but also for other calculations.

In this forecast calculated from the last sale date, it is important to have a measure that returns this date.

4 About DAX measures…
Measure details

« AC + FC » measure (1/2)

This measurement illustrates the forecast calculation.
Note: The last point of sale (actual measurement) is shown to create a continuous line between the actual and the graph; this is more of an aesthetic choice.

```
AC + FC (TB_003) =
VAR First_Date_Selected_To_Forecast = SELECTEDVALUE('Table_EoP To Show'[Start Calculation])
VAR Last_Date_Selected_To_Forecast = SELECTEDVALUE('Table_EoP To Show'[End Calculation/Start Forecast])

VAR Table_To_Forecast=
FILTER(
    SELECTCOLUMNS(
        DATESBETWEEN('Dim Date'[Date],First_Date_Selected_To_Forecast,Last_Date_Selected_To_Forecast),
        "DateSelected", 'Dim Date'[Date],
        "AC", [Actual]
    ),
(
    NOT(ISBLANK([AC]))
    )
)
```

This filter is important for the median calculation method, because the MEDIANX function takes BLANK cells into account. See Patou Tips #13 (Understand difference between Average and Median Calculation).

In the first part of the formula we use the dates from the previous formula, the dates from the "EoP" table are the references for the "Dim Date" table.

Use case with PowerBI
"Finance Toolbox"

Performance of an investment (with CAGR)

4

About DAX measures…
Measure details

« Last Year Good for Forecast » measure

```
VAR Value_By_Day_To_Forecast =
        SWITCH(TRUE(),
            SELECTEDVALUE('Table_Forecast Calculation Method'[Id])=10,AVERAGEX(Table_To_Forecast,[AC]),
            SELECTEDVALUE('Table_Forecast Calculation Method'[Id])=20,MEDIANX(Table_To_Forecast,[AC]))

VAR Forecast =
    SUMX(VALUES('Dim Date'),
        CALCULATE(
          IF(SELECTEDVALUE('Dim Date'[RelativeDay])>0,
            Value_By_Day_To_Forecast)))

VAR Actual =
    SUMX(VALUES('Dim Date'),
        CALCULATE(
          IF(SELECTEDVALUE('Dim Date'[RelativeDay])<=0,
            [Actual])))

RETURN Actual+Forecast
```

This filter with the "RelativeDay" offer a good precision. See Patou Tips #02 Patou Tips (The importance of the Date)

The two variable "Actual" and "Forecast" have the same structure of calculation to have the same context. The combinaison of SUMX, VALUES and CALCULATE functions works easily.

5

About visualizations…
Find the main colors

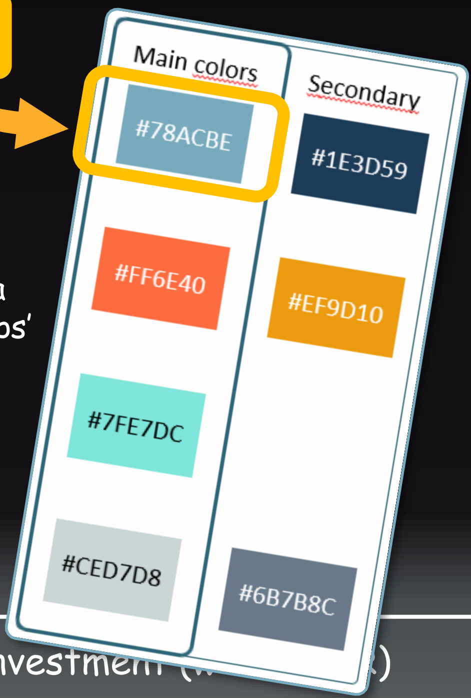
For my book “Story of a point”, I took my inspiration from the book "Super Graphic" by Tim Leong (June 2013). An amazing book for color research.

When I find my main colors, then I search for the hexadecimal code with my smartphone and the "What a color?" mobile application. This application is really usefull to catch the color and the hexadecimal code.

With PowerPoint I test a lot of colors until I find 3 main colors (but here 4) and 3 secondary colors. My favorite color for the book theme and for PowerBI files is the color called:

“Dark Pastel Blue (hexadecimal code: #78ACBE).

Get hexadecimal code with “what a color?” mobile apps’



Use case with PowerBI

“Finance Toolbox”

Performance of an investmen

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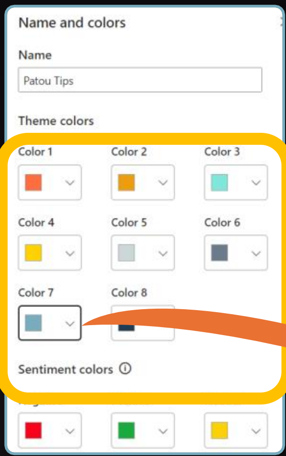
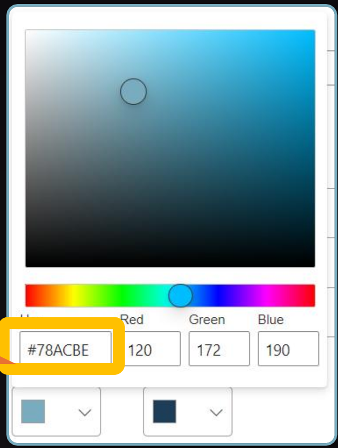
About visualizations...
Create colors theme in PowerBI

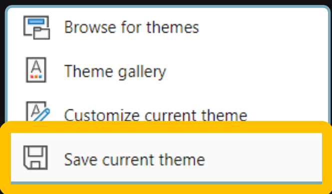
On PowerBI click on "Customize current theme"
Path: View > Themes

Here it's the creation of the 7 colors that I choosed before. And I put the hexadecimal codes for each colors.





Save your final theme, if you want to use it for an another PowerBi project. A json file will be create.



Use case with PowerBI
"Finance Toolbox"

Performance of an investment (with CAGR)

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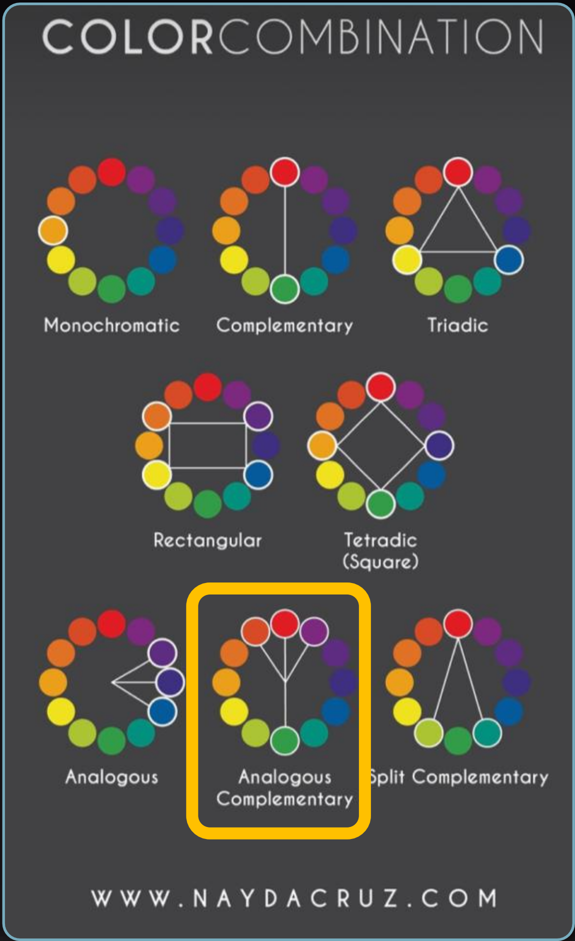
About visualizations...
Find the good harmony



Anything is possible, but often it's better to create your color theme according to color combinationrules.

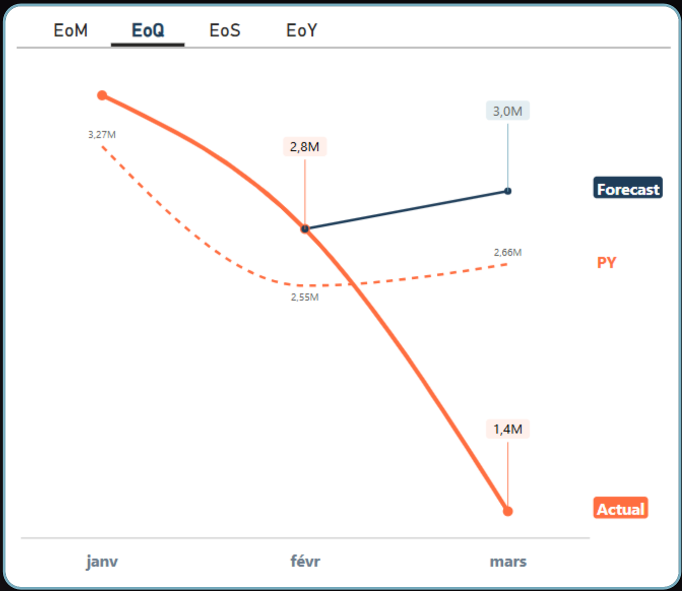
Sometimes, when I find colors, I reopen my project a few days later just to see if the impact or mood I wanted is still the same.

Sometimes I change the theme 4 or 5 times!

The combination that I chose is nearly of the "analogous complementary" harmony.

5

About visualizations...
Line chart with scenario



More details on the PowerBI file (pbix). See link on the second page or in the comment of the LinkedIn post.

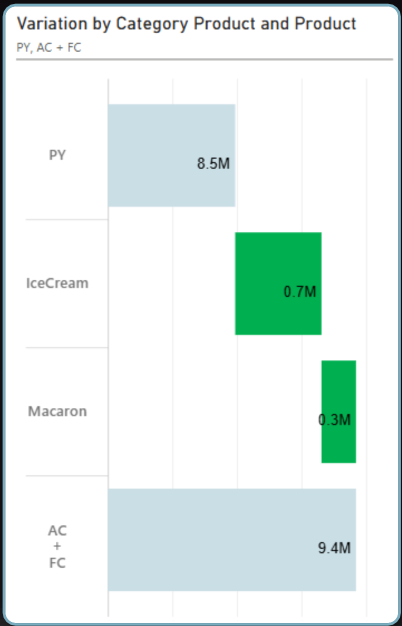
More details on the book "Story of a Point / Financial forecast with PowerBI"

Use case with PowerBI
"Finance Toolbox"

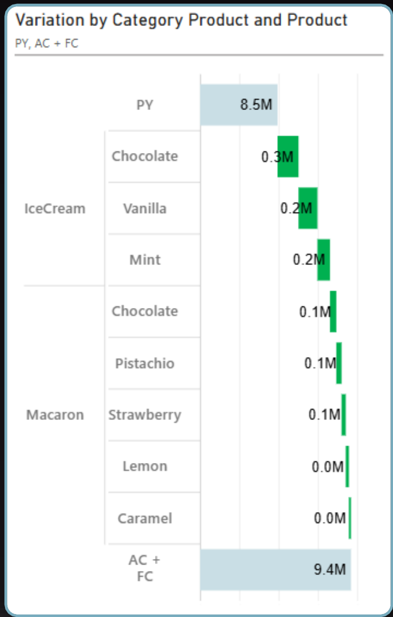
Performance of an investment (with CAGR)

5 About visualizations…
Simple waterfall (1/2)

For this graph, I use the "Simple Waterfall" from "Datanomy Technologies Limited."
This Apps for PowerBI visuals, which can be downloaded from the AppSource, is free, PowerBI certified, is drillable, advanced formatting and fully customizable.



Drill-down



Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

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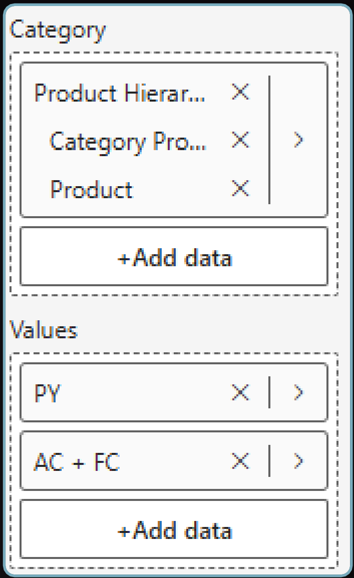
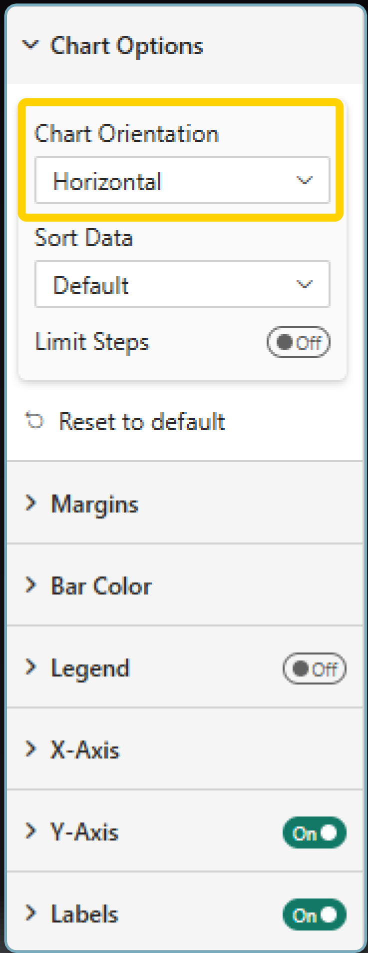
About visualizations...
Simple Waterfall (2/2)





Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

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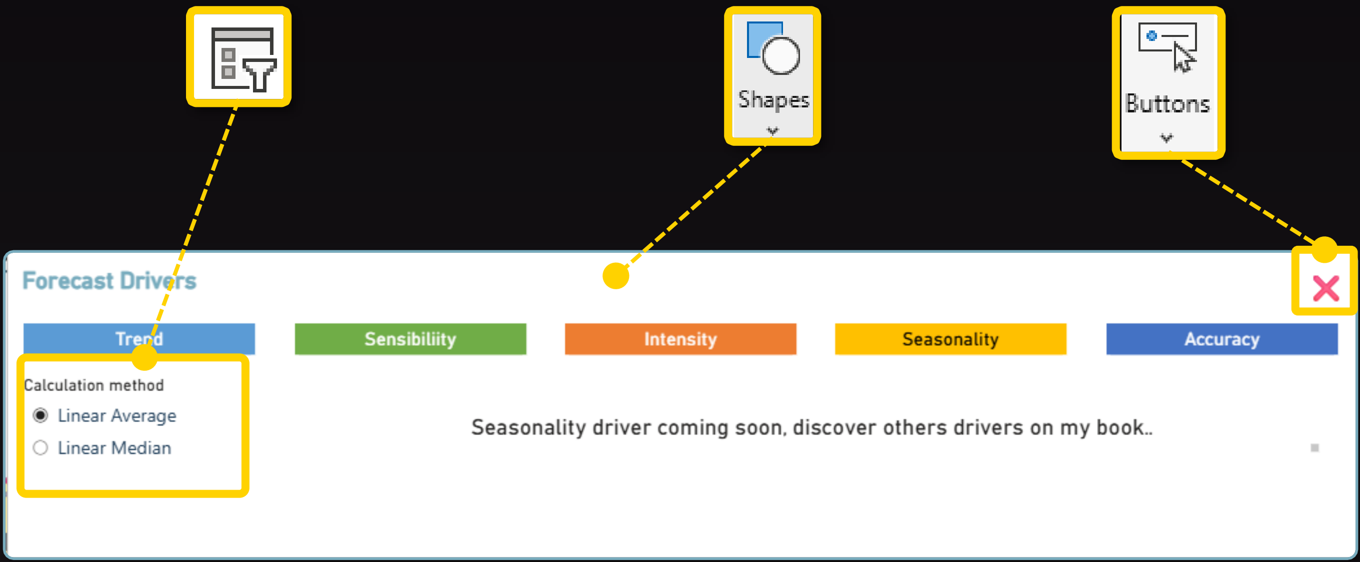
About visualizations...

UI/UX: Create Pop-Up (1/4)

Pop-ups are truly one of the best practices for giving your users the freedom to adjust their work. This creates interactivity and interest in a PowerBI project.



Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

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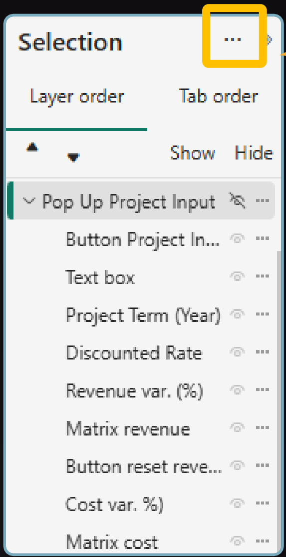
About visualizations...

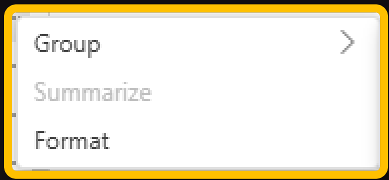
UI/UX: Create Pop-Up (2/4)

Organise objets with "Selection panel"

This step is important to better manage the different objects in your Pop-Up and also to create interactivity with bookmarks (see next page).

In the selection panel, gather all the objects of a Pop-Up into a group





I prefer to group all the objects of a Pop-Up. It will be more easier to work with the bookmarks.

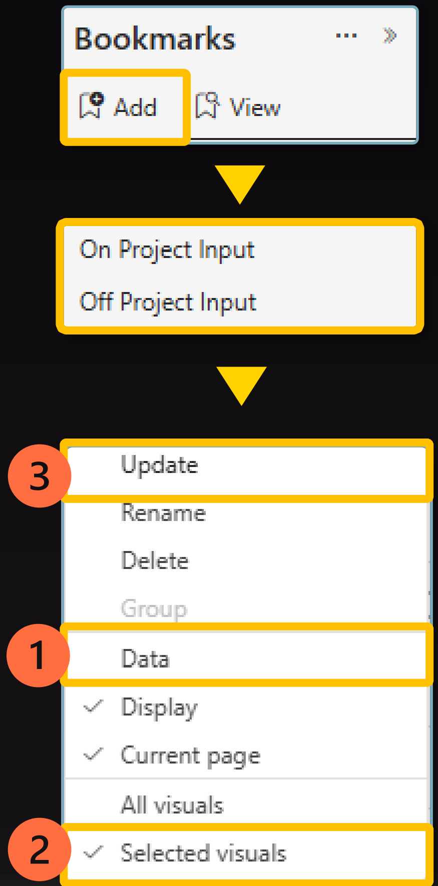
> Here I named it "Pop-Up Project Input"

5

About visualizations...

UI/UX: Create Pop-Up (3/4)

Create “Bookmarks” for the Pop-Up



In the Bookmarks panel, add two bookmarks:

> Create one to show the pop-up (On) and one to close it (Off).

For each Bookmark,

Before all, select the group of object in the selection Pane (see previous page), then:

1 - Unselect “Data”
2 – Select “Selected visuals”
3 – Then “Update”

Use case with PowerBI

“Finance Toolbox”

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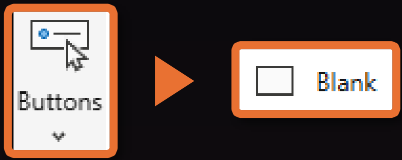
About visualizations…

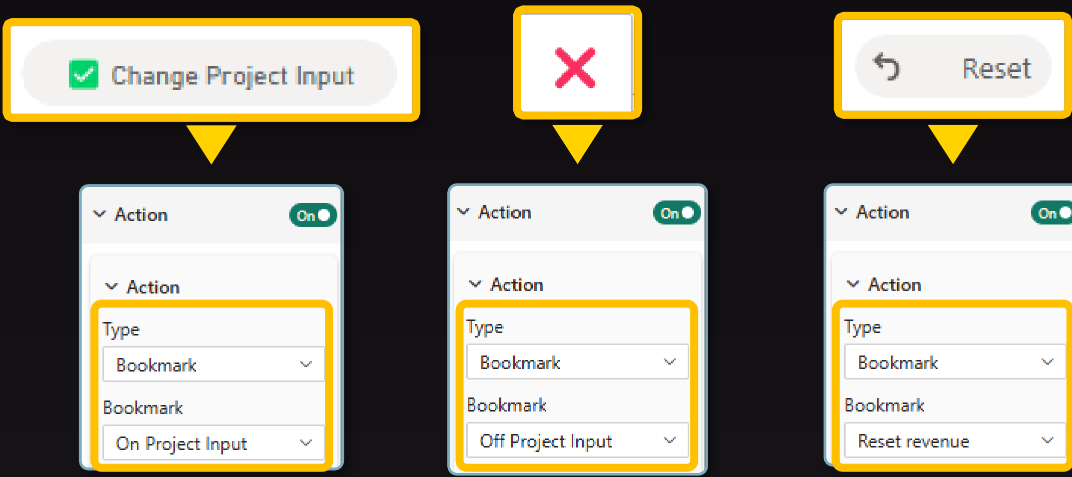
UI/UX: Create Pop-Up (4/4)

Create buttons and their interactivity

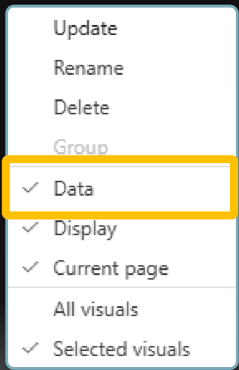
1 Create buttons



2 Create a bookmark for each button (see previous page) and place the bookmark on the "action" configuration button



Don't forget to put on for "data"



Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

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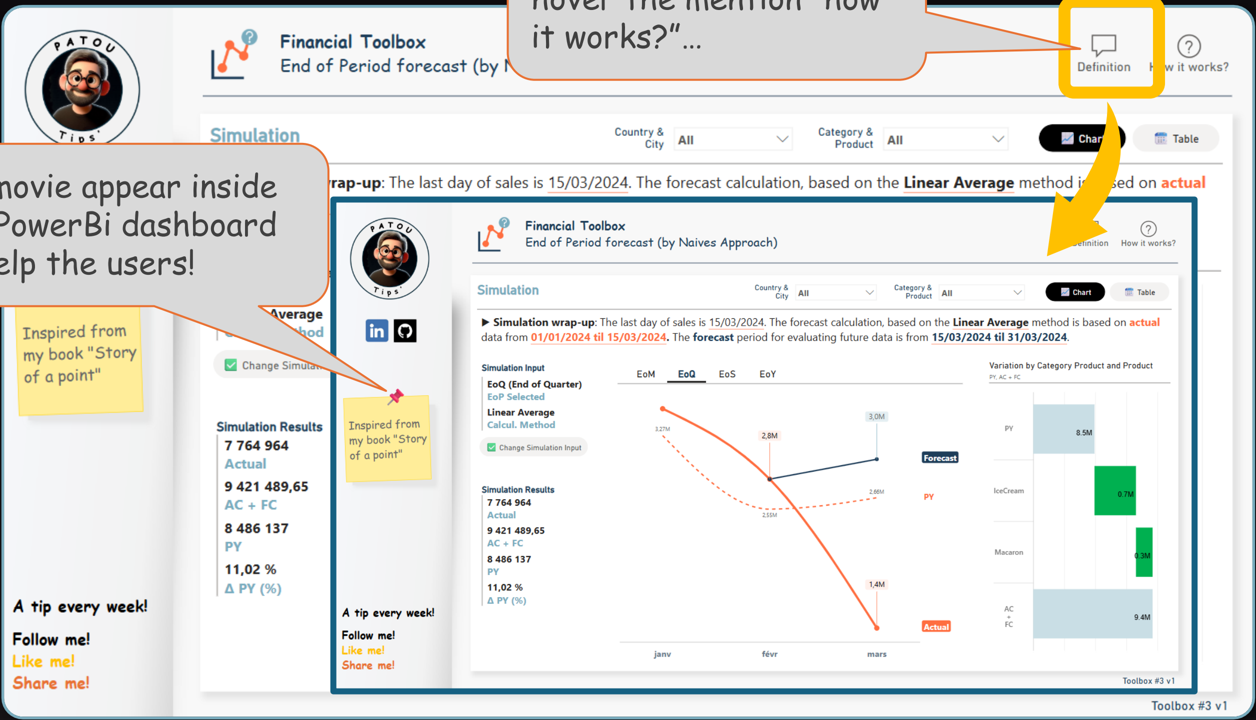
About visualizations…
UI/UX: How it works button (1/4)

Create a movie to help your users in PowerBI

The idea:

When the user drag on hover the mention "how it works?"…



…A movie appear inside the PowerBi dashboard to help the users!

Use case with PowerBI

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Performance of an investment (with CAGR)

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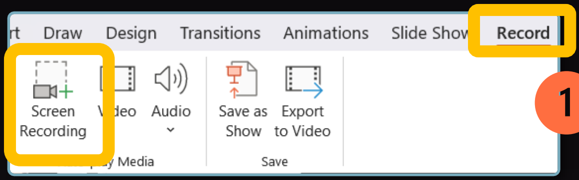
About visualizations…

UI/UX: How it works button (2/4)

Create the movie in PowerPoint



1 On PowerPoint:
Path: Record > Screen Recording

2 A pop-up appear, with the button "Select Area" select the aera on your PowerBi project. A dashed red line indicate you the aera.

3 Click on **record** to start and the touchs ↑ + ⊞ + Q to stop the movie. It's done!

Use case with PowerBI

"Finance Toolbox"

Performance of an investment (with CAGR)

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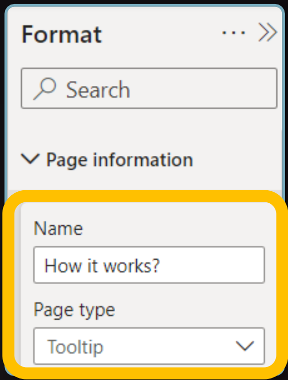
About visualizations...

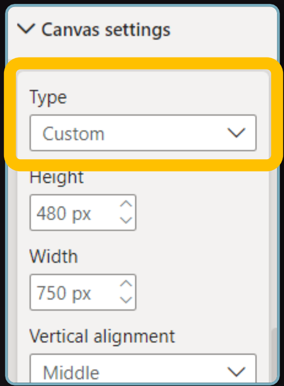
UI/UX: How it works button (3/4)

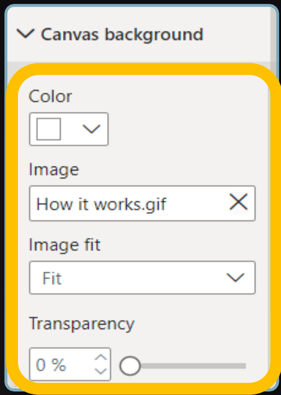
Save the movie as an animated GIF

On PowerBI, create a new page for the animated GIF

Settings of the page



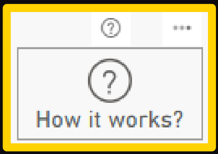




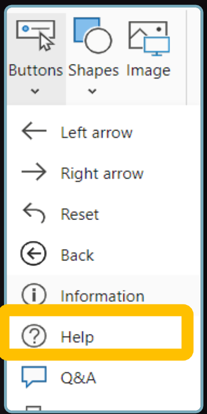
5 About visualizations...

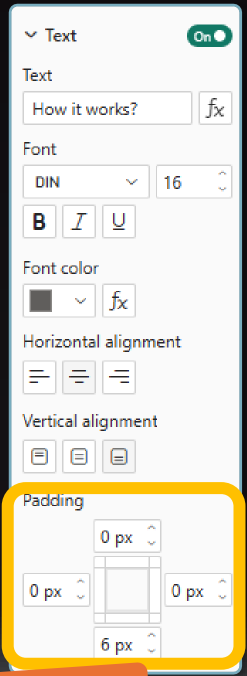
UI/UX: How it works button (4/4)

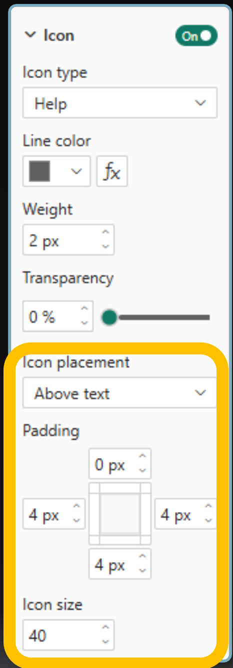
On PowerBI, create a button

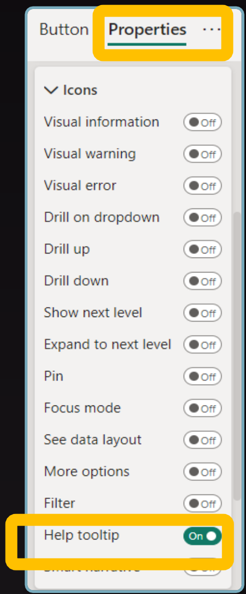


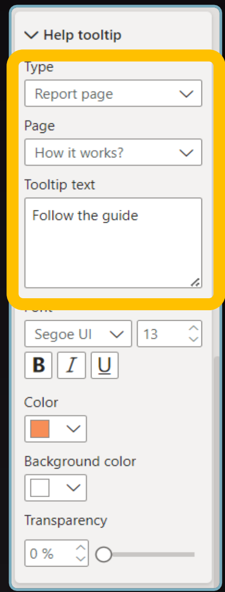
The final result: Drag on hover the button "How it works?" a help symbol (?) appear, To drag on hover this symbol will display the movie.











Use case with PowerBI

"Finance Toolbox"

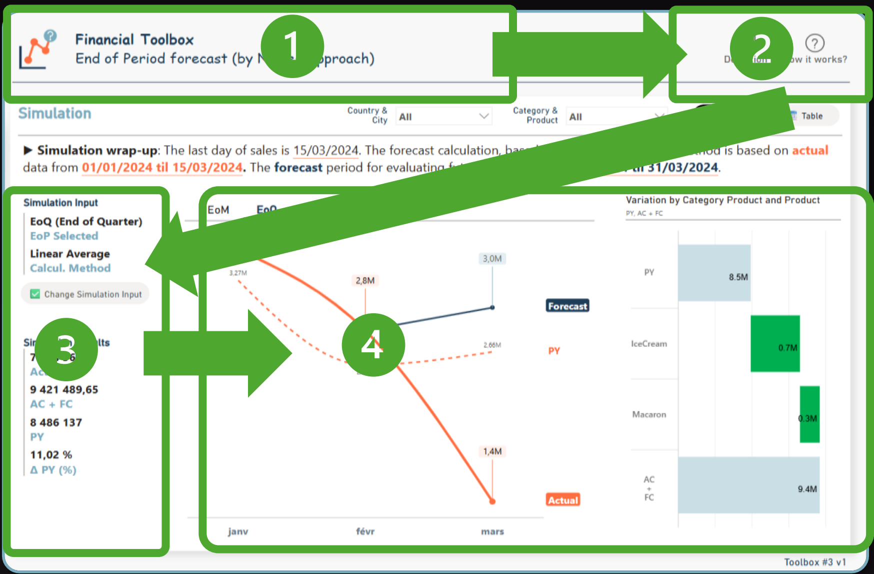
Performance of an investment (with CAGR)

5

About visualizations…

Storytelling: "Z reading pattern".

The organization and prioritization are with the "Z reading pattern". The natural reading direction is often from left to right and from top to bottom in a "Z" shape.



1 Topic of the projet

2 Help and definition

3 KPI of the project

4 Visualization of the project

Use case with PowerBI

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Performance of an investment (with CAGR)

Don’t forget!
This isn't the truth, it's just my truth!

Patou Tips

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